

UNIVERSITY OF EDINBURGH

INTERNSHIP REPORT

M1 INFORMATIQUE CONCEPTS ET APPLICATIONS

Varieties of Semiring Solvers via Multicategorical and Algebraic Constructions

Intern:

Amaury MAZOYER
ENS de Lyon, France

Supervisor :

Ohad KAMMAR
University of Edinburgh,
United Kingdom

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Introduction

- motivation → algebraic simplification is a pain to do manually in dependently typed languages/in proof assistants

- solvers in other languages (Rocq (Grégoire & Mahboubi, 2005), Agda (Kidney, 2019))

read article a little bit

- state of the art: frex debut (Yallop et al., 2018), idris frex (Allais et al., 2025) and involution construction (Allais et al., 2023)

- presentation of the contributions:

- finishing Ohad’s construction proof and correcting the needed hypothesis
- proof of freeness of the construction
- formalising the distributive combination theory in Idris
- implementing the distributive combination semantics
- implementing some free algebras
- testing and comparison

1 Preliminaries

FREX is based on some concepts of *universal algebra*, which gives generic descriptions of algebraic structures and relations between them (Burris & Sankappanavar, 1981, Chp. II). We’ll summarise the relevant concepts, which have been formalised in Idris2 (Allais et al., 2025). We’ll use the example of monoids to illustrate these concepts along the way. We’ll also explain how the simplification works, and we’ll define the distributive combination of theories formally.

find a good arrangement for the subsections

1.1 Syntax

A *finitary signature* $\Sigma = (\text{op } \Sigma, \text{arity})$ consists of a set $\text{op } \Sigma$ of *operators*, and an assignment $\text{arity} : \text{op } \Sigma \rightarrow \mathbb{N}$ associating to each operator in $\text{op } \Sigma$ its *arity*. The usual signature Σ_{Mon} of monoids consists of two operators $(\cdot)$ and 1 , with respective arities 2 and 0. We’ll write $\Sigma_{\text{Mon}} := \{(\cdot) : 2, 1 : 0\}$.

An *algebra* $A = (\underline{A}, A \llbracket - \rrbracket)$ over a signature Σ consists of a set $\underline{A}$ called the *carrier* of A , and a function $A \llbracket f \rrbracket : \underline{A}^n \rightarrow \underline{A}$ called the *interpretation* of f in A for each operator $f : n \in \Sigma$. This gives the semantics to the algebraic language determined by the signature. There may be different algebras for a given signature and carrier set. One can equip the set $\mathbb{N}$ of natural numbers with the monoid structure $(\mathbb{N}, (+), 0)$, $(\mathbb{N}, (\cdot), 0)$ or even $(\mathbb{N}, \max, 0)$.

Given a set X of variables and a signature Σ , the set of Σ -terms over X , denoted $\text{Term}_\Sigma(X)$, is defined by induction:

$$\frac{x \in X}{\text{var } x \in \text{Term}_\Sigma(X)} \quad \frac{t_i \in \text{Term}_\Sigma(X) \quad f : n \in \Sigma}{f(t_1, \dots, t_n) \in \text{Term}_\Sigma(X)}$$

We’ll usually write $x \in \text{Term}_\Sigma(X)$ instead of $\text{var } x \in \text{Term}_\Sigma(X)$. We define *equations in context* $X \vdash l = r$ as triples (X, l, r) where X is a set of variables and l, r are terms in context X called the *left-hand-side* (LHS) and *right-hand-side* (RHS). The associativity equation in Σ_{Mon} is then defined as $x, y, z \vdash x \cdot (y \cdot z) = (x \cdot y) \cdot z$.

An *environment* ρ for a context X , or *X-environment*, in an algebra A is a function $\rho : X \rightarrow \underline{A}$ that assigns to each variable in X an element of the carrier of A . Given a term $t \in \text{Term}_\Sigma(X)$, denoted $X \vdash t$, the algebra A gives an *interpretation* function $A[[t]] : \underline{A}^X \rightarrow \underline{A}$. For an X -environment ρ , $A[[t]]\rho$ is defined by induction:

$$A[[\text{var } x]]\rho := \rho x \quad A[[\text{op}(t_1, \dots, t_n)]]\rho := A[[\text{op}]](A[[t_1]]\rho, \dots, A[[t_n]]\rho)$$

93 An equation $X \vdash l = r$ between terms in $\text{Term}_\Sigma(X)$ is *valid* in an algebra A if for every
 94 environment $\rho : X \rightarrow \underline{A}$, we have $A[[l]]\rho = A[[r]]\rho$. We write this $A \models (X \vdash l = r)$.
 95 For example, the Σ_{Mon} algebras presented so far validate the associativity axiom, but
 96 interpreting the binary operation as subtraction over the integers $(\mathbb{Z}, (-), 0)$ does not:
 97 taking the environment $\{x \mapsto 0, y \mapsto 0, z \mapsto 1\}$, we have $0 - (0 - 1) = 1 \neq -1 = (0 - 0) - 1$.

98 A *presentation* (or *theory*) $\mathcal{T} = \langle \Sigma_{\mathcal{T}}, \text{Ax}_{\mathcal{T}} \rangle$ consists of a signature $\Sigma_{\mathcal{T}}$ and a set $\text{Ax}_{\mathcal{T}}$ of
 99 Σ -equations in context. A $\mathcal{T}$ -*model* A is a $\Sigma_{\mathcal{T}}$ algebra validating all of the axioms in $\text{Ax}_{\mathcal{T}}$.
 100 The **Monoid** presentation consists of the associativity axiom paired with the neutrality
 101 axioms $x \vdash x \cdot 1 = x$ and $x \vdash 1 \cdot x = x$ over the signature Σ_{Mon} .

Let $A = (\underline{A}, A[[_]])$ and $B = (\underline{B}, B[[_]])$ be algebras over the same signature Σ . A
homomorphism of Σ -algebras from A to B is a function $h : \underline{A} \rightarrow \underline{B}$ such that for every
 operator $f : n \in \Sigma$ and every $a_1, \dots, a_n \in \underline{A}$ we have:

$$B[[f]](h(a_1), \dots, h(a_n)) = h(A[[f]](a_1, \dots, a_n))$$

102 In other words, a homomorphism is a semantics-preserving function between the carriers
 103 of the algebras. For example, complex conjugation is an homomorphism between the
 104 Σ_{Mon} -algebras over the complex numbers, since $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$, $\overline{0} = 0$ and $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$,
 105 $\overline{1} = 1$. A *homomorphism* of $\mathcal{T}$ -models is a homomorphism of $\Sigma_{\mathcal{T}}$ -algebras between the
 106 underlying algebras of the models.

1.2 Free models/algebras

108 Let $\mathcal{T}$ be a presentation and X a set of variables. A $\mathcal{T}$ -model over X is just a pair $\langle A, e \rangle$
 109 where A is a $\mathcal{T}$ -model and $e : X \rightarrow \underline{A}$ is an X -environment in this algebra. This concept
 110 is used to formalise the free simplification process in §1.4.

111 A *morphism* $h : A \rightarrow B$ between $\mathcal{T}$ -models over X $\langle A, \rho_A \rangle$ and $\langle B, \rho_B \rangle$ is
 112 a $\mathcal{T}$ -model homomorphism that makes the diagram on the right commute. A
 113 *free $\mathcal{T}$ -model over X* is a $\mathcal{T}$ -model $F(X)$ such that for every $\mathcal{T}$ -model A over
 114 X , there exists a unique morphism of $\mathcal{T}$ -models from $F(X)$ to A . We will
 115 often use the term "free algebra", abbreviated "fral", when $\mathcal{T}$ is implicit.
 116 The existence-and-uniqueness property is called the *universal property* of
 117 the free algebra.

$$\begin{array}{ccc} X & \xrightarrow{\rho_A} & \underline{A} \\ & \searrow \rho_B & \downarrow h \\ & & \underline{B} \end{array}$$

maybe
move out
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syntax
subsec-
tion?

add exam-
ple

1.3 Free extensions

119 An *extension* of A by X is a triple $\langle B, h, e \rangle$ where B is a $\mathcal{T}$ -algebra, $h : A \rightarrow B$ is a
 120 $\mathcal{T}$ -homomorphism and $e : X \rightarrow B$ is a function.

121 Let $b_1 = \langle B_1, h_1, e_1 \rangle, b_2 = \langle B_2, h_2, e_2 \rangle$ be two extensions of A by X . A
 122 *morphism of extensions* $h : b_1 \rightarrow b_2$ is a $\mathcal{T}$ -homomorphism $h : B_1 \rightarrow B_2$ that
 123 makes the diagram on the right commute. An extension $\langle A[X], \text{sta}, \text{dyn} \rangle$ is
 124 *free* when, for every extension $\langle B, h, e \rangle$, there is a unique extension morphism
 125 $[B, h, e] : \langle A[X], \text{sta}, \text{dyn} \rangle \rightarrow \langle B, h, e \rangle$. Free extension is abbreviated "frex",
 126 and a free extension of A by X is usually denoted $A[X]$. Thanks to the
 127 universal property, all free extensions of A by X are isomorphic and therefore
 128 we talk about *the* free extension of A by X . Let A be a commutative monoid. Its frex
 129 by a finite set $X = \{x_1, \dots, x_n\}$ of variables can be represented by $A[X] := \underline{A} \times \mathbb{N}^n$. The

$$\begin{array}{ccccc} & & B_1 & & \\ e_1 \nearrow & & \downarrow h & \nwarrow h_1 & \\ X & & & & A \\ e_2 \searrow & & B_2 & \swarrow h_2 & \end{array}$$

remaining structure is defined by:

$$\begin{aligned}
& \langle a, \langle k_{11}, \dots, k_{1n} \rangle \rangle + \langle b, \langle k_{21}, \dots, k_{2n} \rangle \rangle := \langle a + b, \langle k_{11} + k_{21}, \dots, k_{1n} + k_{2n} \rangle \rangle \\
0 & := \langle 0, \langle 0, \dots, 0 \rangle \rangle \quad \text{sta } a := \langle a, \langle 0, \dots, 0 \rangle \rangle \quad \text{dyn } x_i := \left\langle 0, \left\langle 0, \dots, 0, \underset{i}{1}, 0, \dots, 0 \right\rangle \right\rangle \\
[B, h, e] \quad & \langle a, \langle k_1, \dots, k_n \rangle \rangle := h a + k_1 \cdot e x_1 + \dots + k_n \cdot e x_n
\end{aligned}$$

Every presentation $\mathcal{T}$ induces an equivalence relation between terms using the deduction rules of equational logic. Explicitly, we define a *provability* relation $X \vdash l = r$ inductively as in Fig 1.

$$\begin{array}{c}
\frac{\text{ax} \in \text{Ax}_{\mathcal{T}}}{\text{ax}} \quad (\text{axiom}) \quad X \vdash t = t \quad (\text{reflexivity}) \quad \frac{X \vdash l = r}{X \vdash r = l} \quad (\text{symmetry}) \\
\frac{X \vdash l = m \quad X \vdash m = r}{X \vdash l = r} \quad (\text{transitivity}) \quad \frac{X \vdash l = r \quad \theta : X \rightarrow \text{Term}_{\Sigma_{\mathcal{T}}} Y}{Y \vdash l[\theta] = r[\theta]} \quad (\text{substitution}) \\
\frac{\text{for } i = 1, \dots, n : X \vdash l_i = r_i}{X \vdash \text{op}(l_1, \dots, l_n) = \text{op}(r_1, \dots, r_n)} \quad (\text{congruence})
\end{array}$$

Figure 1: Provability

113

114 Allais et al. (2023) proves that there is an isomorphism of extensions

$$115 \quad \langle A[X], \text{sta}, \text{dyn} \rangle \cong \left\langle \left(\text{Term}_{\Sigma_{\mathcal{T}_A}} X \right) / (X \vdash_{\mathcal{T}_A} =), \text{sta}', \text{dyn}' \right\rangle \text{ with } \text{sta}' : \left| \begin{array}{c} A \rightarrow Q \\ a \mapsto [a] \end{array} \right| \text{ and } \text{dyn}' : \left| \begin{array}{c} X \rightarrow Q \\ x \mapsto [\text{var } x] \end{array} \right|.$$

116 Therefore free extensions represent terms modulo semantics.

1.4 How the fral/frex simplification works

118 (Allais et al., 2025; Corbyn et al., 2022)

1.5 The distributive combination of theories

Given two signatures Σ_1 and Σ_2 , we define their coproduct $\Sigma_1 + \Sigma_2$ as the signature with operators the disjoint union of the operators, but retaining their arity:

$$\Sigma_1 + \Sigma_2 = \{ \iota_i f : n \mid (f : n) \in \Sigma_i, i = 1, 2 \}$$

120 Given a Σ_i -term-in-context $n \vdash_{\Sigma_i} t$, we denote by $n \vdash_{\Sigma_1 + \Sigma_2} \iota_i[t]$ the $\Sigma_1 + \Sigma_2$ -term-in-
121 context obtained by applying the injection ι_i to all the operators of t . Similarly, given a
122 Σ_i -equation in context $eq := (n \vdash_{\Sigma_i} t = t')$, we denote by $\iota_i[eq]$ the $\Sigma_1 + \Sigma_2$ -equation in
123 context $n \vdash_{\Sigma_1 + \Sigma_2} \iota_i[t] = \iota_i[t']$.

124 Let $\mathcal{A}$ and $\mathcal{M}$ be two theories. We define the *distributive combination of $\mathcal{M}$ over $\mathcal{A}$*
125 (Hyland & Power, 2006) to be the theory $\mathcal{M} \triangleright \mathcal{A}$ given by:

$$\begin{aligned}
\Sigma_{\mathcal{M} \triangleright \mathcal{A}} &:= \Sigma_{\mathcal{M}} + \Sigma_{\mathcal{A}} \\
\text{Ax}_{\mathcal{M} \triangleright \mathcal{A}} &:= \iota_1[\text{Ax}_{\mathcal{M}}] \cup \iota_2[\text{Ax}_{\mathcal{A}}] \\
&\cup \left\{ \begin{array}{c} \langle x_i \rangle_{\substack{i=1 \\ i \neq i_0}}^n, \mathbf{y} \mid f(x_1, \dots, x_{i_0-1}, s\mathbf{y}, x_{i_0+1}, \dots, x_n) \\ = s \langle f(x_1, \dots, x_{i_0-1}, \mathbf{y}_j, x_{i_0+1}, \dots, x_n) \rangle_{j=1}^{|\mathbf{y}|} \end{array} \mid \begin{array}{c} f : n \in \Sigma_{\mathcal{M}} \\ s : |\mathbf{y}| \in \Sigma_{\mathcal{A}} \end{array} \right\}
\end{aligned}$$

126 Concretely, the signature is the coproduct signature and the axioms are the union of
127 the axioms to which we added distributivity axioms between the operators of $\mathcal{M}$ and $\mathcal{A}$,
128 stating that every operator of $\mathcal{M}$ distributes over every operator of $\mathcal{A}$. We will call $\mathcal{M}$
129 the multiplicative theory, and $\mathcal{A}$ the additive theory. For example, the theory of semirings

130 is the distributive combination of commutative monoids over monoids. The distributivity
 131 axioms state that:

$$\begin{aligned} x, y, z \vdash x \cdot (y + z) &= (x \cdot y) + (x \cdot z) & x, y, z \vdash (x + y) \cdot z &= (x \cdot z) + (y \cdot z) \\ x \vdash x \cdot 0 &= 0 & x \vdash 0 \cdot x &= x \end{aligned}$$

132 2 Categorical approach to the construction of distributive 133 combinations

134 While using FREX does not require knowledge about category theory, FREX uses many
 135 category-theoretic concepts internally. Therefore, the construction of distributive combi-
 136 nations is firstly category-theoretic. However we won't go into details here, as this would
 137 require a course on 2-categories and multi-categories.

138 2.1 Challenge about constructing distributive combinations

139 write correctly once §1.4 is written.

140 Challenge: not only constructing something that works, but something that is usable
 141 (Corbyn et al., 2022). Therefore, no use of quotients or pushouts or other hard to compute
 142 things. Can also not simply use the construction using coproducts described in Allais et
 143 al., 2025, section 4.4.

144 2.2 Initial result about the free algebra for the distributive combination

145 Let $\mathcal{T}$ be a theory. We say that $\langle F_{\mathcal{T}}, \eta^{\mathcal{T}}, \vDash^{\mathcal{T}} \rangle$ is the free algebra construction for $\mathcal{T}$ if
 146 for every set of variables X , $\langle F_{\mathcal{T}}X, \eta_X^{\mathcal{T}} : X \rightarrow F_{\mathcal{T}}X \rangle$ is the free $\mathcal{T}$ -algebra over X . $\vDash^{\mathcal{T}}$
 147 is given by the universal property of the free algebra: for every function $g : X \rightarrow \underline{A}$,
 148 $\vDash^{\mathcal{T}} g : F_{\mathcal{T}}X \rightarrow A$ is a homomorphism.

149 A first result was given by Ohad Kammar (my internship supervisor) about the free
 150 algebra for the distributive combination of a theory over a commutative theory. A theory
 151 $\mathcal{A}$ is commutative iff for every $f : n \in \Sigma_{\mathcal{A}}$ and every $g : m \in \Sigma_{\mathcal{A}}$,

$$f \left(\begin{pmatrix} g(x_{11}, \dots, x_{1m}) \\ \vdots \\ g(x_{n1}, \dots, x_{nm}) \end{pmatrix} \right) = g \left(f \left(\begin{pmatrix} x_{11} \\ \vdots \\ x_{n1} \end{pmatrix} \right) \quad \cdots \quad f \left(\begin{pmatrix} x_{1m} \\ \vdots \\ x_{nm} \end{pmatrix} \right) \right) \quad (1)$$

152 For monoids, this states that

$$(x \cdot y) \cdot (z \cdot w) = (x \cdot z) \cdot (y \cdot w) \quad 0 \cdot 0 = 0 \quad 0 = 0$$

153 The theory of commutative monoids is therefore a commutative theory, fortunately. Note
 154 that every operator of arity 0 and 1 commutes with itself.

Theorem 1 (Free algebra for the distributive combination, false)

Let $\mathcal{A}$ be a commutative theory, and $\mathcal{M}$ be any theory. Suppose that we have
 $(F_{\mathcal{A}}, \eta^{\mathcal{A}}, \vDash^{\mathcal{A}})$ and $(F_{\mathcal{M}}, \eta^{\mathcal{M}}, \vDash^{\mathcal{M}})$ the free algebra constructions for $\mathcal{A}$ and $\mathcal{M}$.
 We can construct the free algebra construction $(F_{\mathcal{M} \triangleright \mathcal{A}}, \eta^{\mathcal{M} \triangleright \mathcal{A}}, \vDash^{\mathcal{M} \triangleright \mathcal{A}})$ for the dis-
 tributive combination $\mathcal{M} \triangleright \mathcal{A}$ using this data. The construction is concrete, as defined
 in §2.1.

reformulate

155 Even though the construction itself was well-defined, the proof wasn't complete and that
 156 for a good reason: the statement is incorrect. The culprit is distributive laws, or more so
 157 the absence of distributive laws under some hypothesis.

2.3 Relation with distributive laws and why the first result fails

Distributive laws (Beck, 1969) is a concept between monads in the category-theoretical sense. In order to link distributive laws and monads to our problem, we will have to define them properly. We will assume basic knowledge about category theory, mainly about functors, natural transformations and adjunctions. One can refer itself to MacLane (1971).

A *monad* on a category $\mathcal{C}$ is a triple (T, η, μ) where $T : \mathcal{C} \rightarrow \mathcal{C}$ is a functor, $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is a natural transformation called the unit, $\mu : T^2 \rightarrow T$ is a natural transformation called the multiplication, and such that the following diagrams commute:

$$\begin{array}{ccc} T & \xrightarrow{T\eta} & T^2 \\ \eta T \downarrow & \searrow & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array} \quad \begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \mu T \downarrow & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array}$$

Every adjunction $\langle F, G, \eta, \varepsilon \rangle$ gives rise to a monad (MacLane, 1971), by defining $T := GF$, $\eta := \eta$ and $\mu := G\varepsilon F$.

Given two monads (T, η^T, μ^T) and (S, η^S, μ^S) on a category $\mathcal{C}$, a *distributive law* of T over S is a natural transformation $\lambda : TS \rightarrow ST$ such that the following diagrams commute:

$$\begin{array}{ccc} & T & \\ T\eta^S \swarrow & & \searrow \eta^S T \\ TS & \xrightarrow{\lambda} & ST \end{array} \quad \begin{array}{ccc} & S & \\ \eta^T S \swarrow & & \searrow S\eta^T \\ TS & \xrightarrow{\lambda} & ST \end{array}$$

$$\begin{array}{ccccc} T^2 S & \xrightarrow{T\lambda} & TST & \xrightarrow{\lambda T} & ST^2 \\ \mu^T S \downarrow & & & & \downarrow S\mu^T \\ TS & \xrightarrow{\lambda} & ST & & \end{array} \quad \begin{array}{ccccc} TS^2 & \xrightarrow{\lambda S} & STS & \xrightarrow{S\lambda} & S^2 T \\ T\mu^S \downarrow & & & & \downarrow \mu^{ST} \\ TS & \xrightarrow{\lambda} & ST & & \end{array}$$

This law induces a composite monad ST on $\mathcal{C}$ with unit $\eta^{ST} = \eta^S \eta^T$ and multiplication $STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu^S \mu^T} ST$.

Theorem 2 (Zwart and Marsden, 2018)

For a theory $\mathcal{T}$, there is a left adjoint $F^{\mathcal{T}}$ to the obvious forgetful functor $U^{\mathcal{T}} : \mathcal{T}\text{-Alg} \rightarrow \mathbf{Set}$ where $\mathcal{T}\text{-Alg}$ is the category with $\mathcal{T}$ -models as objects and $\mathcal{T}$ -model homomorphisms as morphisms. The functor is defined as follows:

- For a set X , $F^{\mathcal{T}}X := \underline{F_{\mathcal{T}}X}$ is the carrier of the free $\mathcal{T}$ -algebra over X .
- The action on a function $h : X \rightarrow Y$ is defined inductively by:

$$\begin{aligned} F^{\mathcal{T}}(h)(x) &= h(x) \quad \text{for } x \in X \\ F^{\mathcal{T}}(h)(\text{op}(t_1, \dots, t_n)) &= \text{op}(F^{\mathcal{T}}(h)(t_1), \dots, F^{\mathcal{T}}(h)(t_n)) \quad \text{for } \text{op} : n \in \Sigma_{\mathcal{T}} \end{aligned}$$

For a theory $\mathcal{T}$, the *free model monad* induced by $\mathcal{T}$ is then the monad induced by the free/forgetful adjunction $U^{\mathcal{T}} \circ F^{\mathcal{T}}$. We say that a monad T corresponds to a theory $\mathcal{T}$ if T is a free model monad induced by $\mathcal{T}$. It is well know that every finitary monad on the category of sets arises as a free model monad for some algebraic theory.

TODO : explain why this is relevant

Zwart and Marsden (2018) gives multiple results about the absence of distributive laws under some hypothesis. We can regroup these results under 3 categories:

181 $\triangleright$ theorems 3.5, 3.12 and 4.24 all forbid the existence of an idempotent operator in the
 182 multiplicative theory, on top of other conditions.

183 $\triangleright$ theorem 4.7 forbids the existence of two distinct constants in the additive theory.

184 $\triangleright$ theorem 4.17 imposes the abides equation $(a * b) * (c * d) = (a * c) * (b * d)$ to hold for
 185 every operator $*$ in the additive theory.

186 The second and third point are both taken care of by the hypothesis of commutativity
 187 of the additive theory in Theorem 1. Two constants commute with each other if they are
 188 equal, and the abides equation is the commutativity hypothesis for $f = g = *$.

189 However the first point is not accounted for. Under the current hypothesis, idempotent
 190 operators are allowed. Therefore, we need to strengthen our hypothesis.

191 2.4 Corrected result

192 We strengthen the hypothesis on $\mathcal{M}$ by requiring it to be an affine theory. A theory $\mathcal{M}$
 193 is *affine* if every axiom is an equation between terms where each variable occurs at most
 194 once. This allows weakening/deletion of variables, but not contraction. For example,
 195 $x, y \vdash x + y = y + x$ or $x \vdash x \cdot 0 = 0$ are affine equations, but $x \vdash x \vee x = x$ is not. Semigroups,
 196 monoids, groups and their commutative variants are all affine theories, whereas semilattices
 197 are not. This effectively covers the first point of the previous section.

198 We now state the corrected theorem.

Theorem 3 (Free algebra for the distributive combination)

Let $\mathcal{A}$ be a commutative theory, and $\mathcal{M}$ be an affine theory. Suppose that we have
 $(F_{\mathcal{A}}, \eta^{\mathcal{A}}, \models^{\mathcal{A}})$ and $(F_{\mathcal{M}}, \eta^{\mathcal{M}}, \models^{\mathcal{M}})$ the free algebra constructions for $\mathcal{A}$ and $\mathcal{M}$.
 We can construct the free algebra construction $(F_{\mathcal{M} \triangleright \mathcal{A}}, \eta^{\mathcal{M} \triangleright \mathcal{A}}, \models^{\mathcal{M} \triangleright \mathcal{A}})$ for the dis-
 tributive combination $\mathcal{M} \triangleright \mathcal{A}$ using this data. The construction is concrete, as defined
 in §2.1.

reformulate

199 The actual construction uses operads, translations from operads to theories, 2-categories
 200 and multicategories. Therefore, we won't present it nor the proof as is. We are gonna
 201 explain the gist of it.

202 First, start by noticing that a model for the distributive combination is made of a
 203 $\mathcal{M}$ -model and a $\mathcal{A}$ -model on the same carrier set (this ensures that the respective axioms
 204 hold) where moreover the operations of $\mathcal{M}$ distribute over the operations of $\mathcal{A}$. Let X be a
 205 set of variables. The construction start by taking the free $\mathcal{M}$ -algebra over X , $A_0 := F_{\mathcal{M}}X$.
 206 We then take the free $\mathcal{A}$ -algebra over $F_{\mathcal{M}}X$, $A_1 := F_{\mathcal{A}}(F_{\mathcal{M}}X)$. This gives us our candidate
 207 $\mathcal{A}$ -model. But we now need to define the semantics of the operators of $\mathcal{M}$ on the carrier
 208 A_1 . To do this, we "lift" the $\mathcal{M}$ -operators so that the lifted operators on the new carrier
 209 distribute over the $\mathcal{A}$ -operators. Through some complex reasoning and with the help of
 210 the affine hypothesis, we prove that the $\mathcal{M}$ -axioms hold. The commutativity hypothesis
 211 of $\mathcal{A}$ is used to prove the existence of an adjunction between multicategories, which is used
 212 to lift the $\mathcal{M}$ -operators.

213 This construction is concrete: it is solely built from applying the free algebra construc-
 214 tions and some 2-functors in well-chosen categories. The lifted operators can be defined
 215 concretely.

216 2.5 An attempt at constructing the free extension for the distributive 217 combination

218 The natural next step is wanting to extend this result to free extensions. However the
 219 task is not so easy.

220 Not easy does not mean impossible. As [Allais et al. \(2025\)](#) and [Yallop et al. \(2018\)](#)
221 state, the free extension of an algebra A by X is the category-theoretic coproduct of A
222 with the free $\mathcal{A}$ -algebra over X . However, on top of this construction not being practical,
223 it doesn't use the free extensions for the composite theories to construct the one for the
224 combined theory.

225 We managed to define a construction that uses pushouts and the free extensions of
226 the component theories. While still not computable because of the quotient nature of
227 pushouts, this proves that it is indeed possible to use the free extensions of the composite
228 theories to construct the one for the combined theory.

229 finish this proof in my notes

230 write

231 Failed proof under the same hypothesis as previous section → shows where the con-
232 struction stops working: the construction lacks knowledge regarding the interaction of the
233 concrete and abstract part of the free extension.

234 3 Implementation

235 3.1 The Idris frex infrastructure

236 name section

- 237 • maybe talk about setoids ? (and cite appropriate references)
- 238 • go over some basic definitions used such as `Signature`, `Op`, `Algebra`, `Term`, `SetoidAlgebra`,
239 `Equation`, `Presentation`, `Model`

240 3.2 Implementing the distributive combination

- 241 • explain the choice of representation for the theory of the distributive combination
- 242 • quickly go over the implementation of the semantics (the only real relevant part is
243 `psi` but the definition in itself is atrocious, give type declarations only) → maybe
244 give the definition of `DistributiveCombinationStructure`
- 245 • don't go over the details of the axiom preservation and the freeness proofs (we don't
246 have them anyway, also should we be straightforward about this ?)
- 247 • maybe only give the signature of `FreeDistributiveCombination`

248 3.3 Performance evaluation

- 249 • find interactivity and attention thresholds
- 250 • compare with the performances of the base solvers if possible

251 4 Comparison with existing solvers

252 find better name for section

253 only comparison with ([Grégoire & Mahboubi, 2005](#))

- 254 • quick description of how the `ring` tactic of Rocq is implemented (reflection)

- comparison of the theories supported → Rocq supports commutative semirings and commutative rings only, we support an umbrella of different combination of theories → put some sort of table, or the pseudo-matrix diagram of the first meeting

$$\underbrace{\begin{pmatrix} \text{commutative semigroup} \\ \text{commutative monoid} \\ \text{abelian group} \end{pmatrix}}_{\text{additive theory}} \times \underbrace{\begin{pmatrix} \text{semigroup} \\ \text{monoid} \\ \text{group} \end{pmatrix} \times \begin{pmatrix} \text{mere} \\ \text{commutative} \end{pmatrix} \times \begin{pmatrix} \text{mere} \\ \text{involutive} \\ \text{non-reversing involutive} \end{pmatrix}}_{\text{multiplicative theory}}$$

- pros of the tactic over the fral/frex approach (support for additional equations)
- comparison of capabilities (I believe that the `ring` tactic is not complete, whereas the frex approach is) → limitations of adding equations to the tactic, with the example of the involution
To check: I think that you can only add equations that use the know operations, i.e `+`, `*`, `0`, `1`.
- comparison in usability (runtime) → frex might be slower as typechecking takes a long time.

TODO: runtime comparison

Conclusion

Prospects: adapting this work to the construction of the free extension for the distributive combination

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297 **A Internship context**

298 **B Source code**

299 **Acknowledgments**